COSC 39: Theory of Computation

Credit Statement: Talked to Carson Bates'27, and Kiran Jones'27.

Problem 1. Find a fooling set of infinite size for each of the following languages, thus showing that all of them are non-regular:

1. $L_1 := \{0^m 1^n : n \text{ divides } m \text{ and } n, m > 0\}$
2. $L_2 := \{ww : w \in \{0,1\}^\star\}$
3. $L_3 := \{ww^R : w \in \Sigma^\star\}$ (These are even-length palindromes. A palindrome is a string that reads the same forward and backward.)

Solution.

1. Define the fooling set as $F := \{0^m \mid m \in \mathbb{N}\}$. Then for two arbitrary prefixes $0^i, 0^j \in F$ with $i < j$, let the distinguishing suffix be 1^j. Then we have,
   - $0^i 1^j \notin L_1$, because i is smaller than j and hence is not divisible.
   - $0^j 1^j \in L_1$, because j divides j.

   This implies that F is a fooling set of infinite size, and thus L_1 is not regular.

2. Define the fooling set as $F := \{0^m 1 \mid m \in \mathbb{N}\}$. Then for two arbitrary prefixes $0^i 1, 0^j 1 \in F$ with $i \neq j$, let the distinguishing suffix be $0^j 1$. Then we have,
   - $0^i 1 0^j 1 \notin L_2$ since $i \neq j$ we can't break the string into two equal substrings.
   - $0^j 1 0^j 1 \in L_2$ with $w = 0^j 1$.

   This implies that F is a fooling set of infinite size, and thus L_2 is not regular.

3. Define the fooling set as $F := \{0^m 11 \mid m \in \mathbb{N}\}$, Then for two arbitrary prefixes $0^i 11, 0^j 11 \in F$ with $i \neq j$, let the distinguishing suffix be 0^j. Then we have,
   - $0^i 110^j \notin L_3$ since $i \neq j$ we can't break the string into w and w^R.
   - $0^j 110^j \in L_3$ with $w = 0^j 1$.

   This implies that F is a fooling set of infinite size, and thus L_3 is not regular.